

LABORATORY RECORD

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Course	Full Stack Web Development
Experiment	HTML5 Canvas and CSS Styled Webpage

1. TITLE

HTML5 Canvas and CSS Styled Webpage

2. AIM

To create an interactive webpage using HTML5 Canvas and to design a styled webpage using internal CSS with navigation, content cards, buttons, hover effects, and footer.

3. OBJECTIVES

- To create and style an HTML5 canvas element.
- To draw a rectangle, circle, line, and text using JavaScript canvas methods.
- To apply internal CSS for webpage layout and visual styling.
- To create a navigation bar with hover effects.
- To design three content cards using Flexbox.
- To style buttons and footer using CSS.

4. THEORY

HTML5 Canvas provides a drawing area that can be controlled using JavaScript. The 2D rendering context allows shapes, lines, colors, and text to be drawn dynamically. CSS (Cascading Style Sheets) is used to control the appearance and layout of HTML elements. In this program, internal CSS is used for backgrounds, typography, spacing, navigation, cards, buttons, hover effects, shadows, and footer styling.

5. PROCEDURE / ALGORITHM

1. Create an HTML document with the required head and body sections.
2. Create Part A with a heading and a canvas element of 500 × 300 pixels.
3. Access the canvas using `document.getElementById()` and obtain its 2D context.
4. Draw the rectangle, circle, line, and student name using JavaScript.
5. Create Part B and define internal CSS rules for the webpage.
6. Create a header, navigation bar, three content cards, buttons, and footer.
7. Use Flexbox to arrange the cards and add hover effects to links and buttons.
8. Save the HTML file and open it in a browser to verify the output.

6. PROGRAM – PART A

```
<!DOCTYPE html>
<html>

<head>
  <b><h1>PART A</h1></b>
  <!-- Title of the webpage -->
  <title>My Styled Webpage</title>

  <style>
    /* Center the heading */
    h1 {
      text-align: center;
    }

    /* Center the canvas */
    canvas {
      display: block;
      margin: 20px auto;
      border: 2px solid black;
    }
  </style>
</head>

<body>

  <!-- Main Heading -->
  <h1>HTML5 Canvas</h1>

  <!-- Canvas Element -->
  <canvas id="myCanvas" width="500" height="300">
    Your browser does not support the canvas element.
  </canvas>

  <!-- JavaScript for Drawing -->
  <script>

    // Access the canvas
    var canvas = document.getElementById("myCanvas");

    // Get 2D drawing context
    var ctx = canvas.getContext("2d");

    // Draw a Rectangle
    ctx.fillStyle = "violet";
    ctx.fillRect(20, 20, 150, 80);

    // Draw a Circle
    ctx.beginPath();
    ctx.arc(280, 60, 40, 0, 2 * Math.PI);
    ctx.fillStyle = "green";
    ctx.fill();

    // Draw a Line
    ctx.beginPath();
    ctx.moveTo(50, 180);
    ctx.lineTo(350, 180);
    ctx.strokeStyle = "red";
    ctx.lineWidth = 3;
    ctx.stroke();

    // Display Name as Text
    ctx.font = "24px Arial";
    ctx.fillStyle = "blue";
    ctx.fillText("Badana Pranathi", 120, 250);

  </script>
```

6. PROGRAM – PART B

```
<hr>
<br>

<b><h1>PART B</h1></b>

<!-- Title -->
<title>My Styled Webpage</title>

<!-- Internal CSS -->
<style>

    /* Body */
    body{
        margin:0;
        font-family:Arial, sans-serif;
        background:linear-gradient(to right,#74ebd5,#ACB6E5);
    }

    /* Header */
    header{
        background-color:#003366;
        color:white;
        text-align:center;
        padding:20px;
        font-size:35px;
        text-shadow:2px 2px 5px black;
    }

    /* Navigation Bar */
    nav{
        background-color:#222;
        text-align:center;
        padding:15px;
    }

    /* Navigation Links */
    nav a{
        color:white;
        text-decoration:none;
        margin:20px;
        font-size:20px;
    }

    /* Hover Effect for Navigation Links */
    nav a:hover{
        color:yellow;
    }

    /* Content Cards */
    .container{
        display:flex;
        justify-content:center;
        gap:30px;
        margin:40px;
    }

    .card{
        background:white;
        width:250px;
        border:2px solid black;
        border-radius:15px;
        padding:20px;
        box-shadow:5px 5px 10px gray;
        text-align:center;
    }

    .card h2{
        font-family:Georgia;
        font-size:24px;
    }

    .card p{
        font-size:18px;
    }

    /* Button */
    button{
        background-color:#007BFF;
        color:white;
        border:none;
        border-radius:8px;
        padding:10px 20px;
        font-size:16px;
    }

    /* Hover Effect for Button */
    button:hover{
        background-color:green;
        cursor:pointer;
    }

    /* Footer */
    footer{
        background-color:#003366;
        color:white;
```

```

        text-align:center;
        padding:15px;
        margin-top:30px;
        text-shadow:1px 1px 3px black;
    }
</style>
</head>
<body>
    <!-- Header -->
    <header>
        My Styled Webpage
    </header>

    <!-- Navigation Bar -->
    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Services</a>
        <a href="#">Contact</a>
    </nav>

    <!-- Three Content Cards -->
    <div class="container">

        <!-- Card 1 -->
        <div class="card">
            <h2>Card 1</h2>
            <p>This is the first content card.</p>
            <button>Read More</button>
        </div>

        <!-- Card 2 -->
        <div class="card">
            <h2>Card 2</h2>
            <p>This is the second content card.</p>
            <button>Read More</button>
        </div>

        <!-- Card 3 -->
        <div class="card">
            <h2>Card 3</h2>
            <p>This is the third content card.</p>
            <button>Read More</button>
        </div>

    </div>

    <!-- Footer -->
    <footer>
        © 2026 My Styled Webpage
    </footer>
</body>
</html>

```

7. CODE EXPLANATION

Element / Property	Purpose
<canvas>	Creates the drawing area for graphics.
getContext("2d")	Obtains the 2D drawing context.
fillRect()	Draws a filled rectangle.
arc()	Creates the path for the circle.
moveTo() / lineTo()	Defines the starting and ending points of a line.
stroke()	Draws the defined line/path.
fillText()	Displays text inside the canvas.
linear-gradient()	Creates a gradient background.
display:flex	Places the cards in a flexible horizontal layout.
border-radius	Creates rounded corners for cards and buttons.
box-shadow	Adds a shadow effect around cards.
:hover	Changes the appearance when the mouse pointer is over an element.
background-color / color	Sets background and text colors.
text-shadow	Adds a shadow effect to text.

8. TEST CASES

Test Case	Action	Expected Result	Status
TC-01	Open HTML file	Webpage loads successfully	Pass
TC-02	View Part A	Canvas displays rectangle, circle, line and name	Pass
TC-03	View navigation	Home, About, Services and Contact links displayed	Pass
TC-04	View cards	Three cards arranged horizontally	Pass
TC-05	Hover over links/buttons	Hover styling is applied	Pass

9. OUTPUT

Output 1 – HTML5 Canvas

HTML5 Canvas Output	Displayed Result
Heading	HTML5 Canvas – centered heading
Rectangle	Violet filled rectangle
Circle	Green filled circle
Line	Red line with width 3
Text	Badana Pranathi displayed in blue

Output 2 – My Styled Webpage

Component	Output
Header	My Styled Webpage with dark header and text shadow
Navigation	Home About Services Contact
Content	Three white rounded cards with shadows
Buttons	Read More buttons with hover effect
Footer	© 2026 My Styled Webpage

10. RESULT

Thus, the HTML5 Canvas and CSS styled webpage was successfully created. The canvas graphics, navigation bar, content cards, buttons, hover effects, and footer were implemented and verified in the browser.